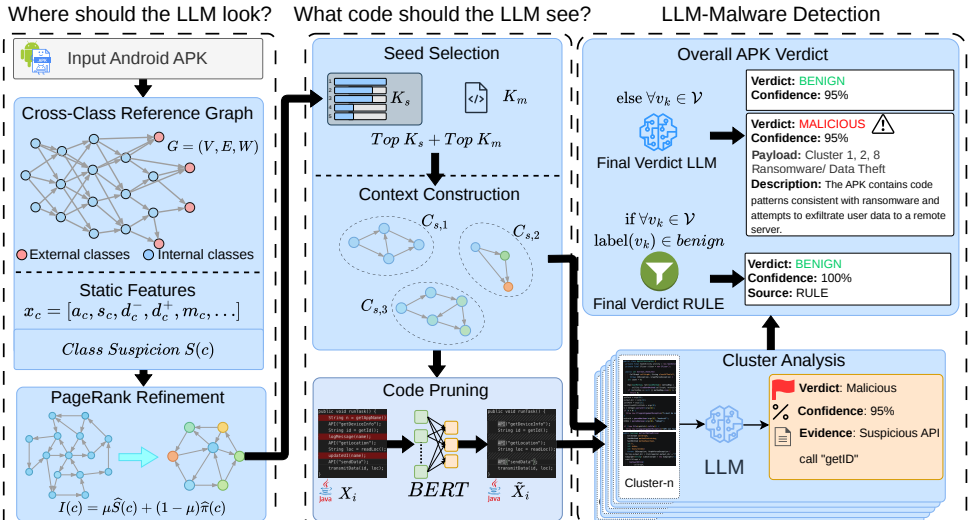

Where should the LLM look?
What code should the LLM see?
LLM-Malware Detection
Input Android APK
Cross-Class Reference Graph
$G = (V, E, W)$
External classes
Internal classes
Static Features
$x_c = [a_c, s_c, d_c^-, d_c^+, m_c, \ldots]$
$Class\ Suspicion\ S(c)$
PageRank Refinement
$I(c) = \mu\widehat{S}(c) + (1-\mu)\widehat{\pi}(c)$
Seed Selection
K_s
K_m
$Top\ K_s + Top\ K_m$
Context Construction
$C_{s,1}$
$C_{s,2}$
$C_{s,3}$
Code Pruning
X_i
BERT
$\tilde{X}_i$
Overall APK Verdict
else $\forall v_k \in \mathcal{V}$
Final Verdict LLM
Verdict: BENIGN
Confidence: 95%
Verdict: MALICIOUS
Confidence: 95%
Payload: Cluster 1, 2, 8
Ransomware/ Data Theft
Description: The APK contains code patterns consistent with ransomware and attempts to exfiltrate user data to a remote server.
if $\forall v_k \in \mathcal{V}$
$\text{label}(v_k) \in benign$
Final Verdict RULE
Verdict: BENIGN
Confidence: 100%
Source: RULE
Cluster Analysis
Cluster-n
LLM
Verdict: Malicious
Confidence: 95%
Evidence: Suspicious API call "getID"